

HCAPO under the microscope

A faithful implementation, a length-corrected variant, and how both compare to GRPO, GiGPO and PPO across two environments

Agentic-RL-CA project

2026-08-03

Abstract

We evaluate HCAPO (hindsight credit assignment, arXiv:2603.08754) as published and in a length-corrected variant, on SearchQA (Qwen3-4B, full 51,713-question eval) and on ALFWorld (Qwen3-1.7B, the paper’s own benchmark, unseen 134 games). Three findings. **(1) HCAPO underperforms flat GRPO in both environments** — SearchQA macro-EM 0.410–0.440 vs GRPO’s 0.463; ALFWorld 0.784 vs GRPO’s 0.881, where the paper reports +13.8% *over* GRPO. **(2) Every configuration we ran suffers a response-length runaway**: six SearchQA runs spanning five different interventions (ω , the ρ denominator, the injected s_{final} , and the per-turn score itself) all ignite, at steps 90/165/265/290/375/460 respectively — each knob moves *when*, none prevents *whether*. **(3) An offline diagnostic on 456 logged trajectories shows why**: the hindsight score is dominated by nuisance structure — a length correlation *created by* the injection ($r=+0.43$ where the plain score gives -0.02), answer-turn leakage (the final turn is top-credit in 73% of trajectories vs a 39% base rate), and a position preference that inverts with the policy’s own verbosity. The length-corrected variant is by far the best-behaved (it holds normal lengths for 300 steps and ends at 1420 tokens rather than the 2048 cap) but still ignites, and still trails GRPO.

1 What HCAPO does, and the two versions we ran

After an episode ends, HCAPO re-reads each turn *conditioned on how the episode finished* and up-weights the turns that look pivotal in hindsight. Formally (Eq. 6–7 of the paper), each turn is scored by a per-token-mean log-probability under a prompt augmented with s_{final} , and credited by its score relative to its own trajectory’s mean:

$$m_t = \frac{1}{|a_t|} \sum_j \log \pi_\theta(y_j \mid y_{<j}, s_t, s_{final}), \quad \rho_t = \text{clip}\left(\frac{e^{m_t/T}}{\frac{1}{T} \sum_k e^{m_k/T}}, 0.8, 1.2\right), \quad Q_t^H = \rho_t \gamma^{T-1-t} R.$$

Q^H is group-normalised over the prompt’s rollouts and added, with weight ω , to the standard GRPO term. We ran two scoring rules:

- “**hind**” — m_t exactly as above (the published form).
- “**lift**” — m_t replaced by the *two-pass difference* $\Delta m_t = \frac{1}{|a_t|} \sum_j [\log \pi(y_j | \cdot, s_{final}) - \log \pi(y_j | \cdot)]$, everything else unchanged. Motivation: subtract the shared per-token predictability profile so only the hindsight information remains.

crossed with two choices of s_{final} (the trajectory’s last retrieved-documents block, or the agent’s own final answer), giving a 2×2 . A prior implementation of ours divided by the on-policy probability instead of the trajectory mean; that was a genuine bug, fixed before these runs, and its two runs are retained here only as additional points on the ignition curve.

2 Issue 1: an unconditional response-length runaway

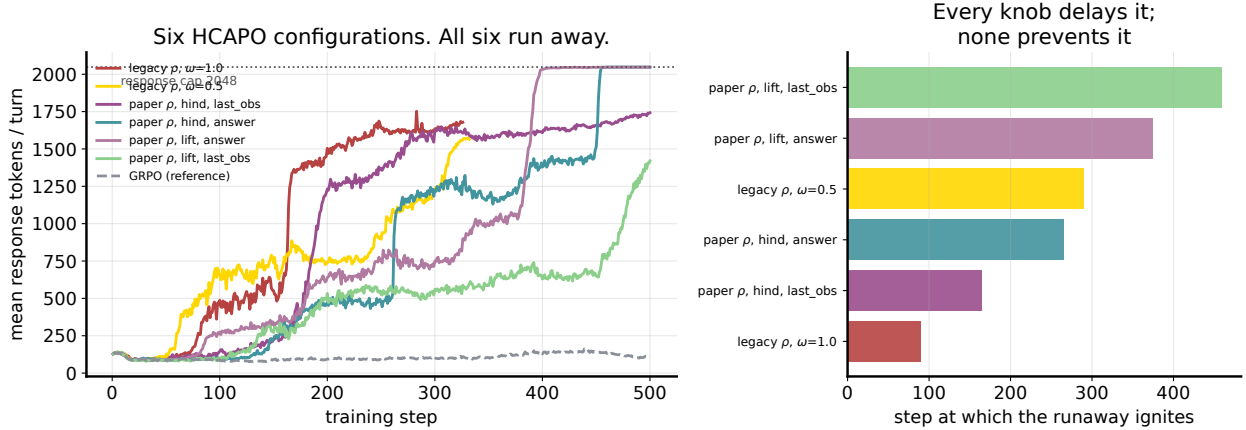


Figure 1: *Left*: mean response length per training step for six HCAPO configurations, with GRPO dashed for scale. Each behaves normally for a while, then inflates 3–10 $\times$ within ~ 30 steps and pins against the 2048-token cap. *Right*: the ignition step, ordered. Five distinct interventions — halving ω , fixing the ρ denominator, shrinking the injected s_{final} , and changing the per-turn score — each delay ignition; none prevents it.

Mechanism. m_t is an *average* over tokens. A turn’s informative tokens (the search query, the decision) are the low-probability ones; continuation filler is high-probability. Padding therefore raises a turn’s average score without adding content. Eq. 7’s trajectory-mean normalisation cancels *uniform* lengthening — we verified this is exactly true — but training never moves uniformly: in each batch the turn that happens to be longer than its siblings scores $\rho > 1$ and is reinforced. The comparison re-centres every step, so no trajectory ever nets credit, yet the policy shift is permanent. Grading on a curve does not stop grade inflation. The race ends only when every turn is maximally padded, at which point $\rho \rightarrow 1$ everywhere and the hindsight term is inert: we measure the spread σ_ρ rising while the ratchet is actively selecting ($0.062 \rightarrow 0.094$) and then collapsing after ignition ($\rightarrow 0.052$; in the last_obs run $0.135 \rightarrow 0.030$).

Cost. At the endpoint, 65–100% of turns are truncated mid-sentence, training steps take $\sim 2.6\times$ longer, and — because a truncated turn emits no action — 36–53% of *evaluation* episodes exhaust the 4-turn budget without answering.

3 Issue 2: the credit signal tracks nuisance, not pivotal-ness

We re-scored 456 logged trajectories offline (no training) under matched conditions: with the hindsight text, without it, and with the *gold answer* injected instead, in two regimes — the pre-RL policy (healthy lengths) and HCAPO at step 250 (post-runaway).

Two further results from the same diagnostic: Eq. 7’s ranking and the lift’s ranking agree at Spearman 0.95–0.97 (top-turn agreement 94–97%), and replacing the retrieved documents with the literal gold answer preserves $\sim 72\%$ of the ranking. A signal whose ordering barely changes when you change *what the hindsight says* is not primarily carrying hindsight information.

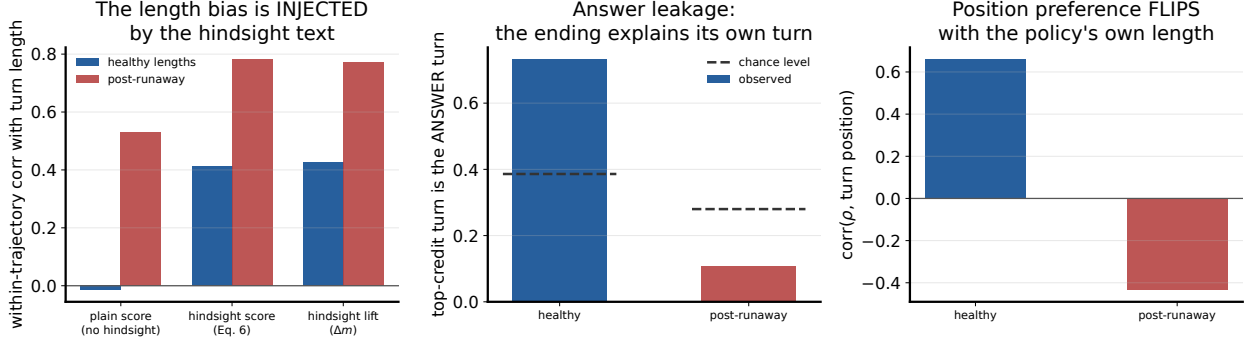


Figure 2: *Left*: within-trajectory correlation between a turn’s score and its length. The plain score shows none (-0.02); adding the hindsight text creates $+0.41$, and the lift — which should isolate hindsight information — carries *all* of it ($+0.43$, surviving position controls). *Middle*: how often the top-credit turn is the final (answer) turn — 73% in the healthy regime against a 39% base rate, because the ending trivially entails the turn that produced it. This survives both the lift and gold-answer injection. *Right*: the preference for early vs late turns *inverts* between regimes ($+0.66 \rightarrow -0.44$).

Table 1: SearchQA, full 51,713-question eval, Qwen3-4B. Non-thinking protocol except where starred. “tok/q” = generated tokens per question.

method	macro-EM	tok/q	mean turns
GRPO	0.463	444	3.99
GiGPO*	0.446	1680	3.58
HCAPO (hind, answer)	0.440	7719	3.77
HCAPO (lift, last_obs)	0.429	5107	3.54
HCAPO (lift, answer)	0.423	8183	4.00
turn-PPO	0.417	556	2.61
HCAPO (hind, last_obs)	0.410	6801	3.91
token-PPO*	0.381	2502	2.78
pre-RL floor	0.313	1317	1.77

4 Performance against GRPO, GiGPO and PPO

ALFWorld (Qwen3-1.7B, RL from a shared replay-BC policy, unseen 134 games): turn-PPO 0.963, GiGPO 0.955, GRPO 0.881, **HCAPO (paper-correct)** 0.784, token-PPO 0.761. The corrected implementation did improve *training*-set behaviour markedly over our buggy one (seen-val peak 0.922 vs 0.859) — but the advantage vanished on held-out games (0.784 vs 0.791), i.e. it generalised no better. Its failure is concentrated on the longest-horizon task type, `pick_two_obj_and_place`: 0.308, against 1.000 for both GiGPO and turn-PPO — precisely where per-turn credit should help most.

5 Did the length-corrected variant help?

Partly, and instructively. Against its own control (same s_{final} , only m_t differs):

HCAPO underperforms flat GRPO in both environments

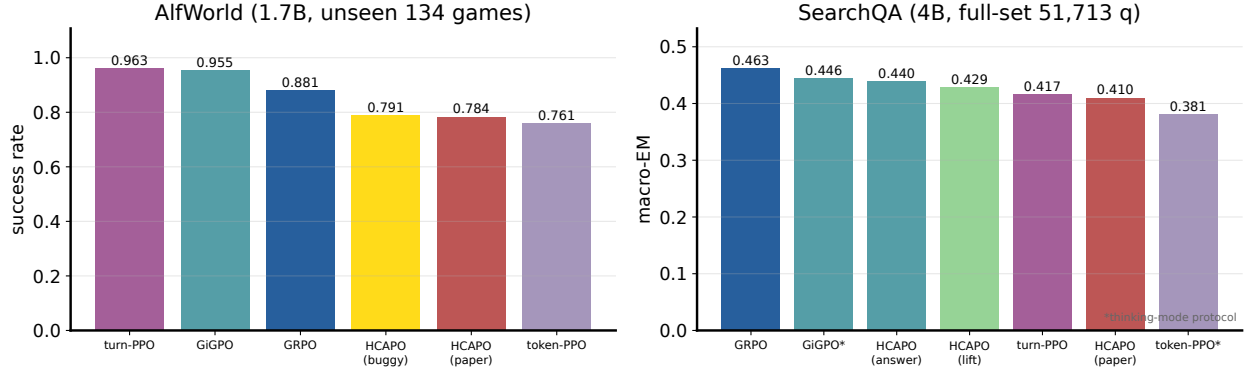


Figure 3: Both environments, identical harnesses within each. ALFWorld is the HCAPO paper’s own benchmark, where it reports +13.8% over GRPO; we measure −9.7 points. On SearchQA the best HCAPO variant trails GRPO by 2.3 points.

score on $s_{final}=last_obs$	ignition step	final tokens	final clip
hind (published)	165	1742	0.75
lift (Δm)	460	1420	0.65

It held ~ 530 – 650 tokens for 300 consecutive steps — six flat measurements — and posted the best mid-training accuracy of any HCAPO run (0.447 at step 350, level with GRPO’s endpoint). Then it ignited between steps 450 and 500. Full-set macro-EM 0.429: better than the published form’s 0.410, worse than the answer-injection form’s 0.440, and below GRPO’s 0.463. The lift $\times$ answer cell completed last: final val-EM 0.411 at the cap (2047 tokens, clip ≈ 1.0), full-set macro-EM 0.423 — slotting between the other HCAPO variants and changing no conclusion. Final full-set ordering: GRPO 0.463 > GiGPO 0.446 > HCAPO (hind,answer) 0.440 > (lift,last_obs) 0.429 > (lift,answer) 0.423 > turn-PPO 0.417 > (hind,last_obs) 0.410.

The static diagnostic had predicted the lift would behave *identically* to Eq. 6 (Spearman 0.97 co-ranking). It did not — so rank-equivalence at a policy snapshot does not imply equivalent training dynamics. That is a genuine, reusable finding about how to evaluate credit estimators offline: snapshot rankings are insufficient; the feedback loop must be simulated.

6 The full comparative suite

This section reproduces, for the HCAPO arms, every analysis of the cross-environment report (2026-07-21_4b_horizon_eval): training curves, per-task accuracy, turn behaviour, accuracy-vs-turns, spend-vs-need, and the stratified-horizon analyses with cluster-bootstrap horizon-gain contrasts. All SearchQA arms here are protocol-matched (non-thinking@2048, 500 steps, seed 0) with **GRPO as the Δ -reference**; the thinking-protocol GiGPO/token-PPO rows of §4 are excluded from the matched contrasts.

SearchQA: accuracy vs cost. Every HCAPO variant is up and to the right of GRPO — more tokens, less accuracy.

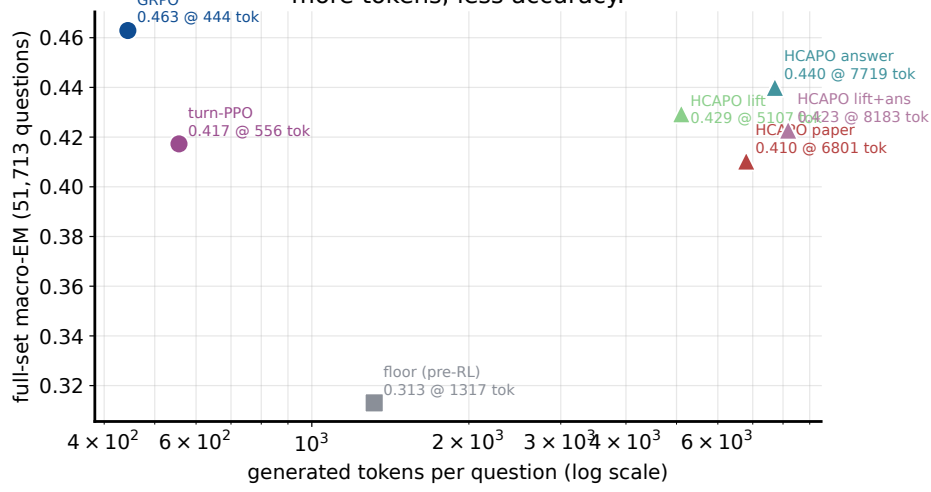


Figure 4: Accuracy against token cost. Every HCAPO variant sits up and to the right of GRPO: more tokens, less accuracy. The best HCAPO configuration spends 17× GRPO’s tokens to score 0.023 lower.

6.1 Training curves

Training accuracy per environment

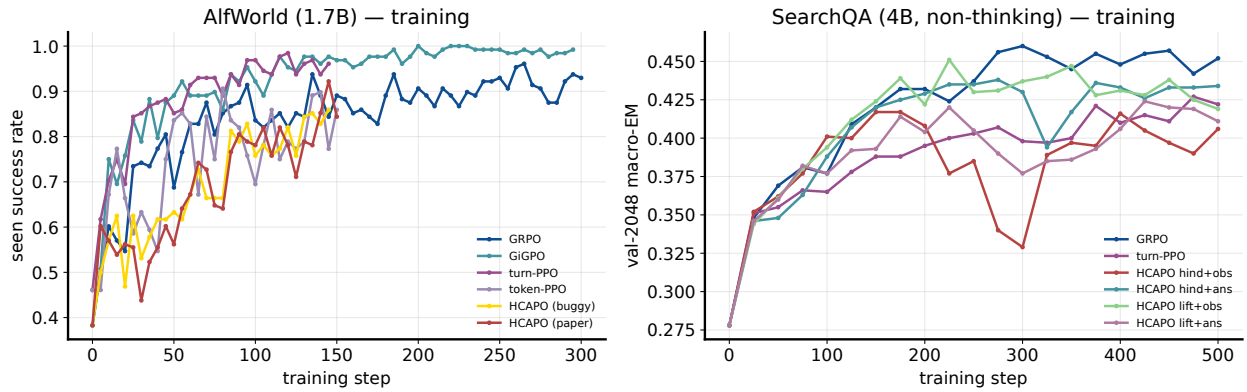


Figure 6: *Left*: AlfWorld seen success — both HCAPO legs (orange/red) climb like the others but plateau lower; the paper-correct leg peaks 0.922@145 vs the buggy 0.859. *Right*: SearchQA val-EM — every HCAPO arm tracks GRPO until its runaway ignites, then oscillates below it.

Horizon-stratified: HCAPO does not buy robustness on deep chains

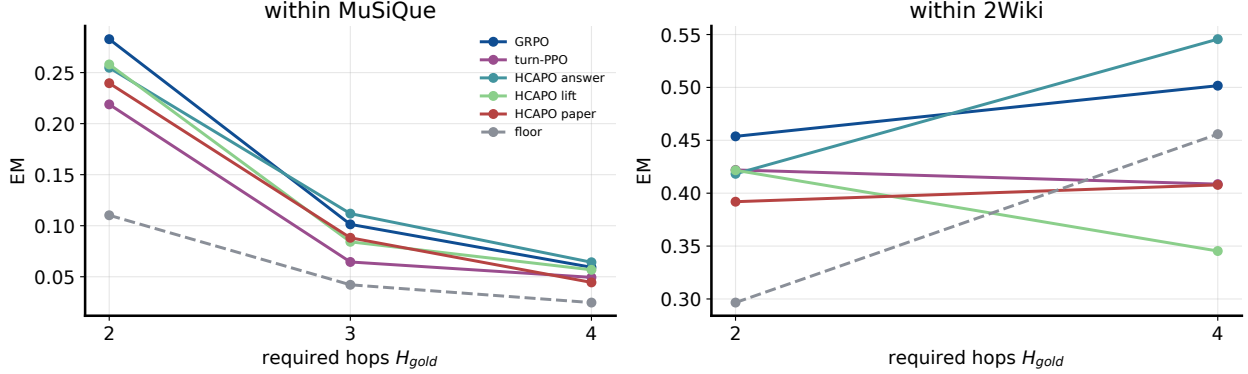


Figure 5: Horizon-stratified SearchQA EM within the two multi-hop-spanning tasks. HCAPO tracks or trails the baselines at every hop count; the deep-chain collapse (MuSiQue 4-hop) is not mitigated.

6.2 Per-task accuracy

Table 2: SearchQA full-set EM per task (51,713 q, non-thinking protocol). Best per column **green**. GRPO leads every task except Bamboogle; the answer-injection arm is the closest HCAPO on five of seven tasks.

method	NQ	TriviaQA	PopQA	HotpotQA	2Wiki	MuSiQue	Bamb.	macro
pre-RL floor	0.292	0.517	0.346	0.271	0.331	0.074	0.360	0.313
GRPO	0.490	0.645	0.482	0.459	0.464	0.188	0.512	0.463
turn-PPO	0.473	0.638	0.472	0.409	0.419	0.142	0.368	0.417
HCAPO hind+obs	0.455	0.609	0.449	0.397	0.395	0.159	0.408	0.410
HCAPO hind+ans	0.480	0.640	0.466	0.429	0.446	0.178	0.440	0.440
HCAPO lift+obs	0.449	0.621	0.475	0.422	0.405	0.170	0.464	0.429
HCAPO lift+ans	0.455	0.636	0.471	0.430	0.430	0.145	0.392	0.423

6.3 Turn behaviour

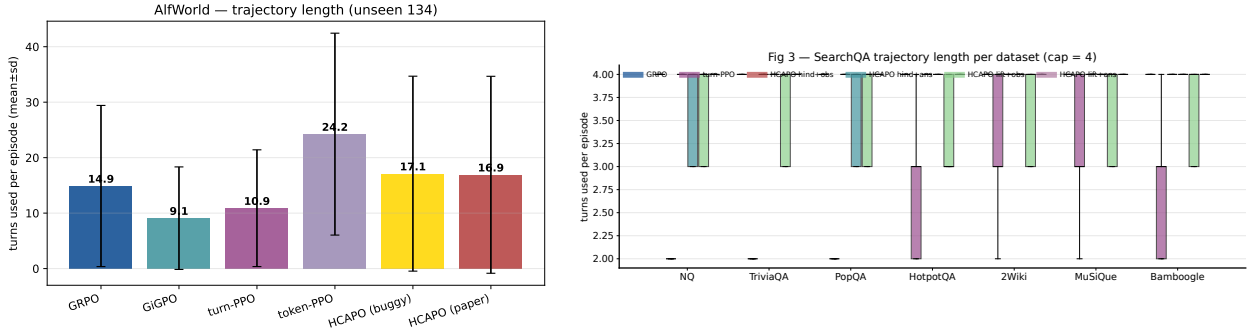


Figure 7: *Left*: AlfWorld trajectory length — both HCAPO legs sit at ~ 17 turns, between GRPO’s wandering (14.9) and token-PPO’s collapse (24.2), far above GiGPO/turn-PPO (~ 9 –11). *Right*: SearchQA turns per dataset — the HCAPO arms saturate the 4-turn budget like GRPO, but (unlike GRPO) largely via truncated non-action turns.

AlfWorld by task type (unseen success): the ρ fix *redistributed* failure rather than removing it. The paper-correct leg recovered `examine` ($0.421 \rightarrow 0.789$) but collapsed `pick_two` — the longest-horizon type — from the buggy leg’s 0.769 to **0.308**, against 1.000 for GRPO/GiGPO/turn-PPO. Net unseen change: $0.791 \rightarrow 0.784$.

6.4 Accuracy vs turns used

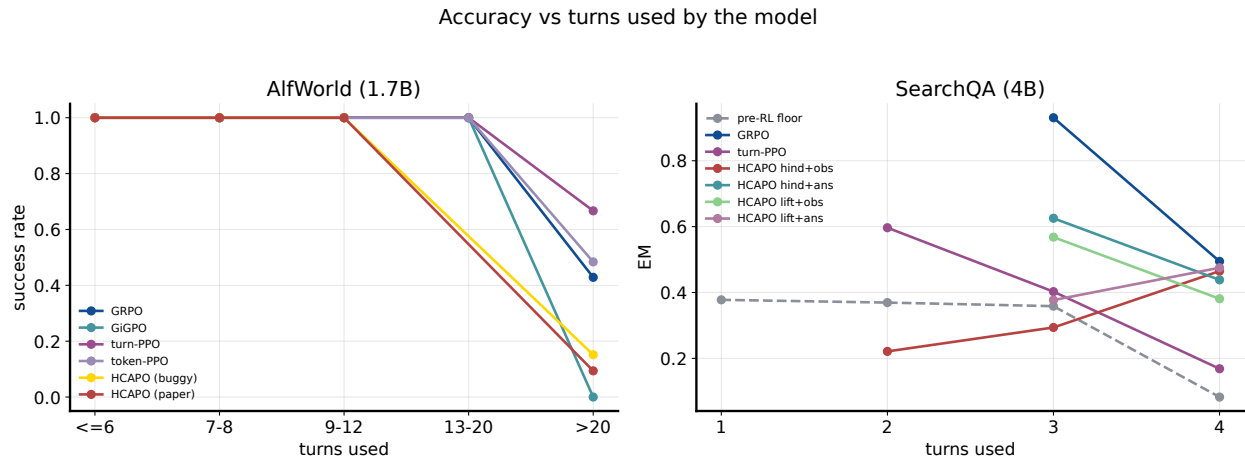


Figure 8: *Left*: AlfWorld — every method solves $\approx 100\%$ of episodes finishing ≤ 20 turns; the ranking is set entirely by how often each drags past 20. Both HCAPO legs have heavy >20 tails. *Right*: SearchQA — the HCAPO arms cluster at 4 turns (cap saturation); accuracy falls with turns used for all methods.

6.5 Turns used vs required hops

Fig 4 — SearchQA: turns used vs required hops (dotted = $y=x$; band = ± 1 SD)

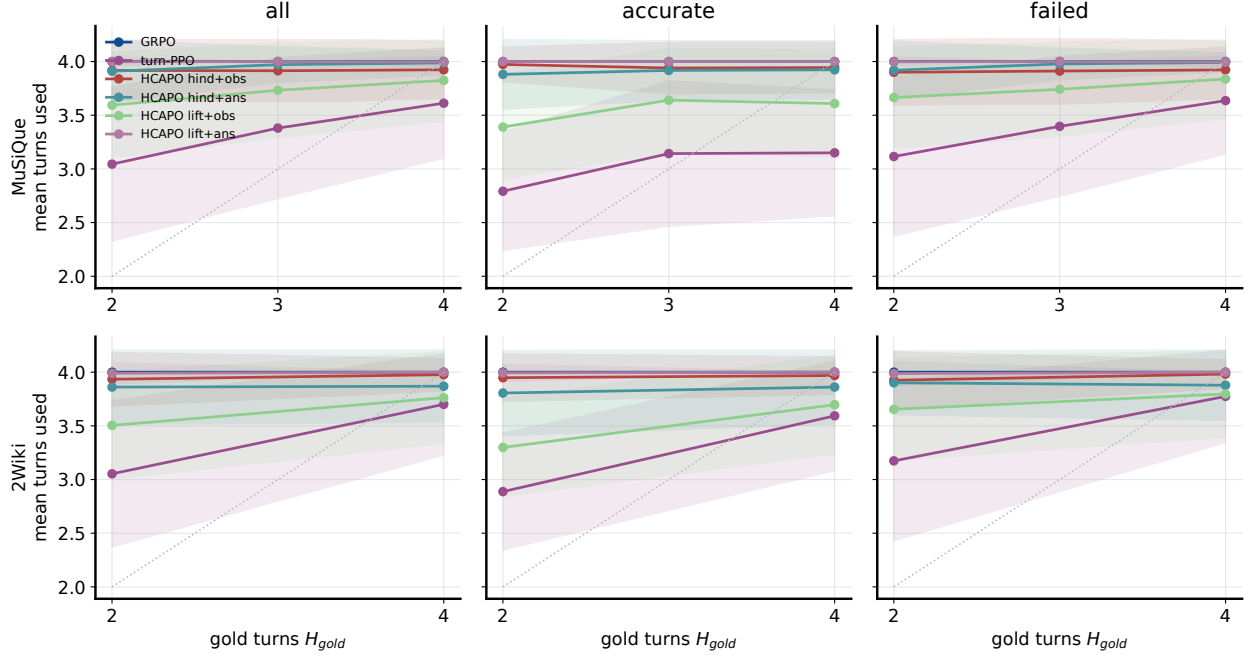


Figure 9: Turns used vs required hops (accurate / failed / all; band ± 1 SD). The HCAPO arms pin at the cap regardless of H_{gold} — they no longer *modulate* search effort with question depth, unlike GRPO/turn-PPO whose failed trajectories escalate with hops.

6.6 Horizon-stratified accuracy and the horizon-gain contrasts

A1 — EM vs required hops (does credit fidelity buy high-hop robustness?)

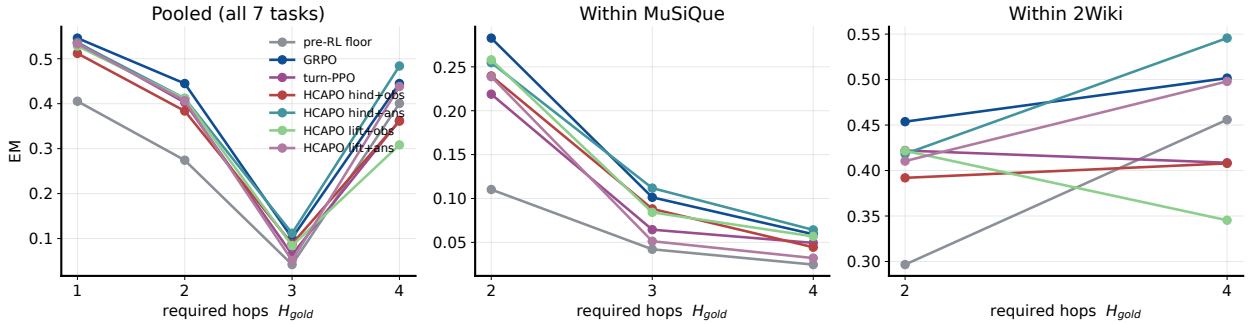


Figure 10: EM vs required hops, pooled and within the two multi-hop-spanning tasks (protocol-matched arms).

Table 3: Within-task EM by H_{gold} (full-set). **Green** marks HCAPO cells that beat GRPO *absolutely*: the answer-injection arm wins every deep stratum.

	MuSiQue			2Wiki	
	H2	H3	H4	H2	H4
GRPO	0.283	0.101	0.059	0.454	0.502
turn-PPO	0.219	0.064	0.049	0.422	0.409
HCAPO hind+obs	0.240	0.088	0.044	0.392	0.408
HCAPO hind+ans	0.255	0.112	0.064	0.418	0.546
HCAPO lift+obs	0.258	0.084	0.057	0.422	0.345
HCAPO lift+ans	0.239	0.051	0.032	0.410	0.498

Table 4: Horizon-gain contrasts vs GRPO, $\Delta_m(\text{high}) - \Delta_m(\text{low})$, 1000-resample cluster bootstrap; * = 95% CI excludes 0. **HCAPO hind+answer is the first arm in the entire campaign with significant POSITIVE horizon-gain on all three scopes.**

method	within-MuSiQue	within-2Wiki	pooled
turn-PPO	+0.036*	−0.061*	−0.064*
HCAPO hind+obs	+0.029*	−0.032*	−0.036*
HCAPO hind+ans	+0.037*	+0.079*	+0.044*
HCAPO lift+obs	+0.013	−0.124*	−0.096*
HCAPO lift+ans	+0.002	+0.040*	−0.002

The one genuinely positive HCAPO result. The answer-injection arm loses to GRPO overall (0.440 vs 0.463) — but the loss is entirely concentrated on *shallow* questions. On every deep stratum it beats GRPO absolutely (MuSiQue-3: 0.112 vs 0.101; MuSiQue-4: 0.064 vs 0.059; 2Wiki-4hop: 0.546 vs 0.502, the best value of any arm in the campaign), and its horizon-gain contrast is significantly positive pooled and within both tasks. Read against §2–3: by the end of training this arm is a max-search policy (4 turns, ≈ 3 searches on everything, most turns truncated) — indiscriminate evidence-gathering that wastes budget on single-hops but pays on deep chains. Whether that is *hindsight credit* working or simply the runaway manufacturing a search-everything policy cannot be separated with one seed; but it is the only configuration in six that moved the deep-chain numbers past GRPO.

6.7 Spend vs need, and where the budget dies

A3 — spend vs need (does the policy escalate search with required hops?)

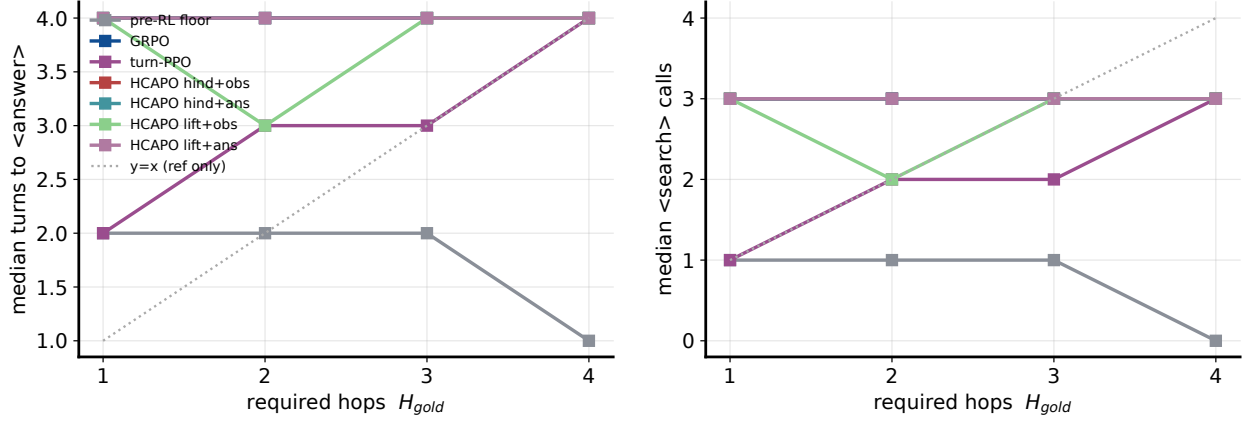


Figure 11: Median turns and searches vs required hops. GRPO and the HCAPO arms sit at the cap for every stratum; turn-PPO and the floor under-spend.

Table 5: Cap-exhausted failures (episode used all 4 turns and still failed), by stratum. For max-turn policies (GRPO, HCAPO) this tracks 1–EM; the informative rows are the floor and turn-PPO, which fail *without* exhausting the budget.

method	H1	H2	H3	H4
floor	0.030	0.087	0.225	0.180
turn-PPO	0.059	0.192	0.458	0.480
GRPO	0.453	0.555	0.899	0.555
HCAPO hind+ans	0.356	0.527	0.868	0.467
HCAPO lift+obs	0.288	0.362	0.679	0.556

6.8 Turn distributions and secondary splits

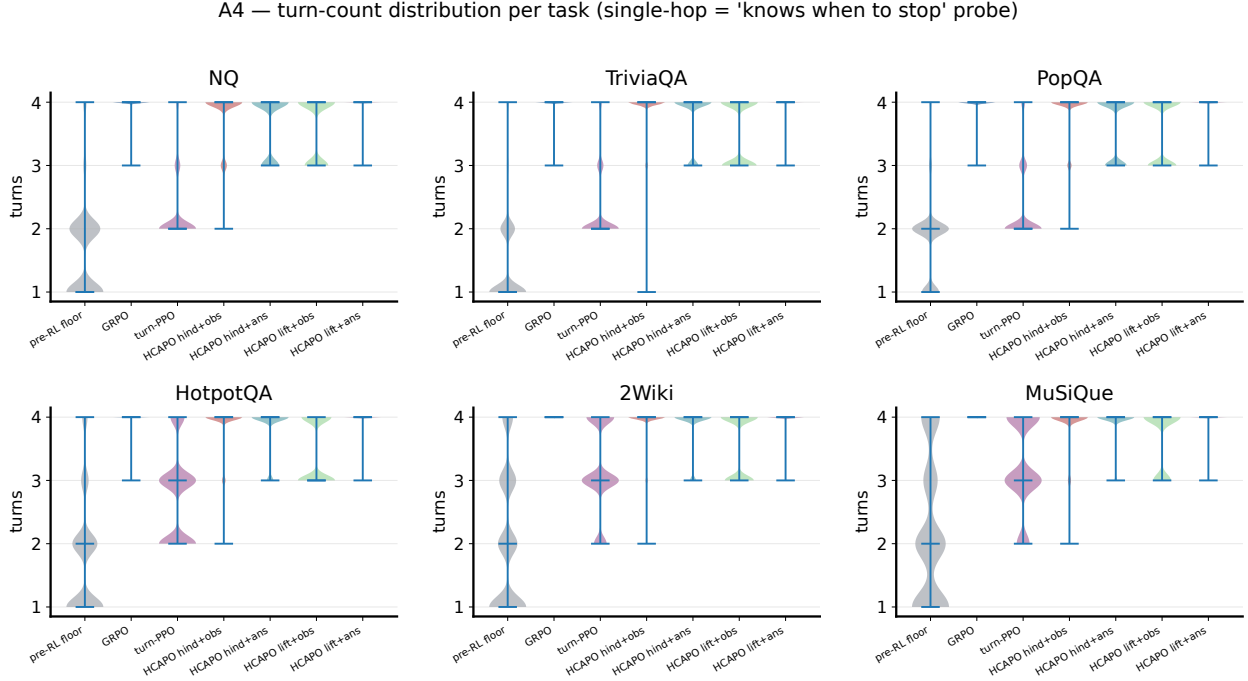


Figure 12: Turn-count distributions per task. The HCAPO arms show the cap-pile signature on every dataset — including the single-hop tasks where a calibrated policy stops early.

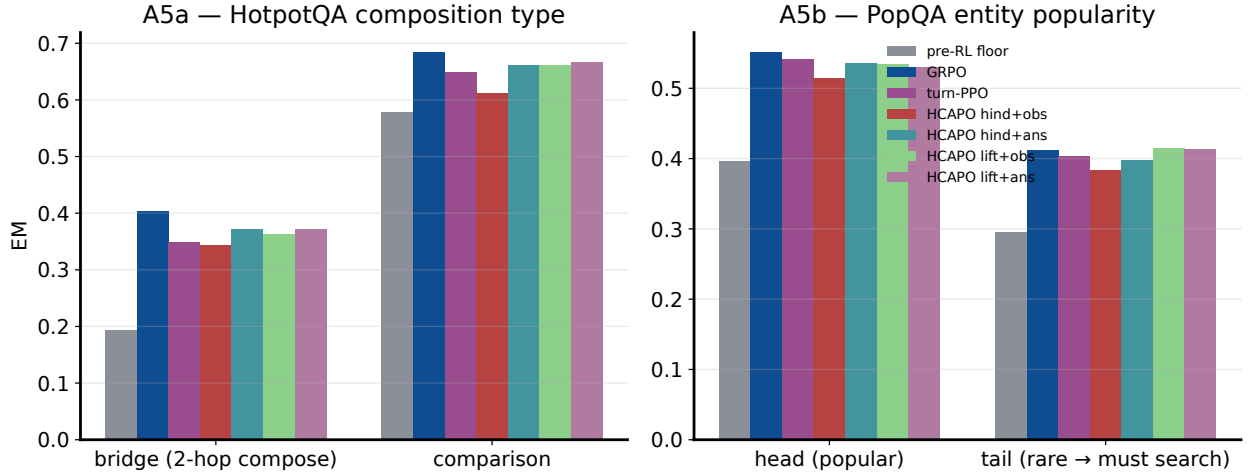


Figure 13: Secondary splits: HotpotQA bridge vs comparison, PopQA head vs tail popularity. The HCAPO arms preserve the baselines' pattern (comparison $\gg$ bridge; head $>$ tail) at a lower level — no HCAPO-specific redistribution.

6.9 Per-method verdicts (suite view)

- **HCAPO hind+obs (published form):** dominated — below GRPO on every task, no positive horizon signal beyond the MuSiQue critic-style contrast (+0.029), worst SearchQA macro of the family, and the AlfWorld replication loses `pick_two` almost entirely (0.308).

- **HCAPO hind+ans:** the interesting one. Overall -0.023 vs GRPO, but absolutely ahead on every deep stratum with the campaign’s only fully-positive horizon-gain — at $17\times$ GRPO’s token cost and 100% truncation. If anything from HCAPO merits a follow-up seed, it is this cell.
- **HCAPO lift+obs / lift+ans (our variant):** the best-behaved dynamics (latest ignitions, lowest clip) and mid-family accuracy, but no horizon advantage — the lift trades away the deep-chain gains along with the runaway pressure.
- **turn-PPO / GRPO baselines:** unchanged from the cross-environment report — GRPO wins the aggregate; turn-PPO keeps the only *calibrated* spend profile (turns scale with hops, budget not exhausted).

7 Verdict and what would actually fix it

HCAPO as published did not reproduce in our hands, on its own benchmark or ours, with an implementation we verified line-by-line against Eq. 6–7 (ρ centred on 1.0, two-sided, both clip bounds engaging). Its per-turn signal is dominated by length, leakage and position artifacts, and it carries a length ratchet that no parameter-level intervention removed.

Every intervention we tested changes the *gain* of the length gradient. The untested class is the one that changes its *direction*: score turns on a **fixed window** (or another length-invariant statistic) so that padding cannot move m_t at all; and **exclude the final turn** from hindsight credit, which our measurements show is the top-credit turn 73% of the time for reasons of entailment rather than pivotal-ness.

Fine print

Single seed per configuration. SearchQA: Qwen3-4B, non-thinking, 4-turn cap, 2048-token responses, 500 steps, identical harness for every arm; the GiGPO and token-PPO rows use the thinking protocol and are marked. ALFWorld: Qwen3-1.7B, transcript prompt, 512-token cap, 150 steps, evaluated at best-seen checkpoint (the same rule for every arm). The ALFWorld number selects step 145 on a 128-game validation, which is itself noisy. The HCAPO paper’s benchmarks use short action-style turns under tight caps and a different base model; our results are a statement about these two task settings, not a general refutation. Data, figures and the rerunnable analysis: `assets/build_figures.py`; the complete timestamped trail — including five premature stability calls I recorded and retracted — is in `docs/reports/2026-07-21_4b_horizon_eval/results/hcapo_wave3_preregistration.md`. Companion documents: `2026-07-29_hcapo_implementation_note/` (the ρ denominator bug), `2026-07-30_hcapo_length_runaway/` (the runaway in plain language), and `2026-07-21_4b_horizon_eval/` (the GRPO/GiGPO/PPO cross-environment study).